

soils4med

SOIL health monitoring and information systems FOR
sustainable soil management in the MEDiterranean region

MedSIS manual *USER Guide*



Authors: Roberto Demontis (demontis@crs4.it), Eva Lorrai (eva@crs4.it), Mais Atiyat (mais.atiyat@hotmail.com)

Delivery dates:

Table of contents:

2. First Access	4
3. User interface	5
The resources list panel	6
Dataset's resource panel	8
Map's resource panel	11
Documents's resource panel	14
Geostory's resource panel	14
Dashboard's resource panel	14
4. How To Requests Data Access	15

This document describes the general use of the SIS platform for a generic user.

The MED-SIS is essentially a catalogue of soil data resources based on Geonode (<https://geonode.org/> see the documentation at <https://docs.geonode.org/en/latest/>).

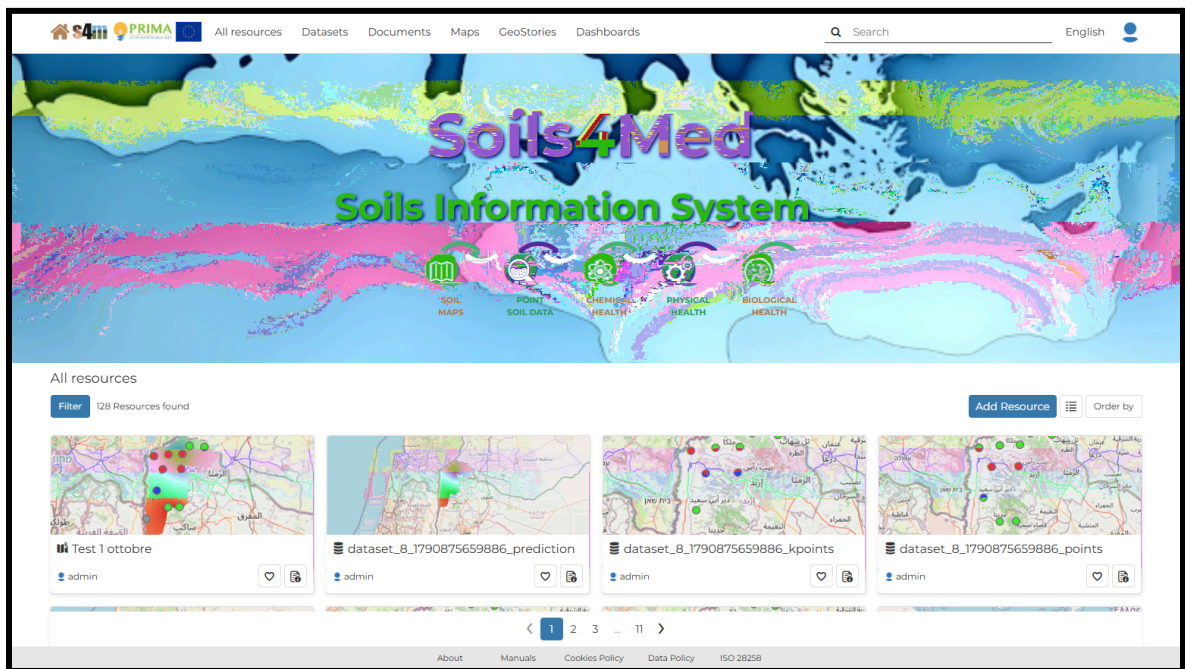


Figure 1: The home page

The MEd-SIS system has a moderate user registration by default; this means that the users must be validated by the staff after the registration (Figure 2) .

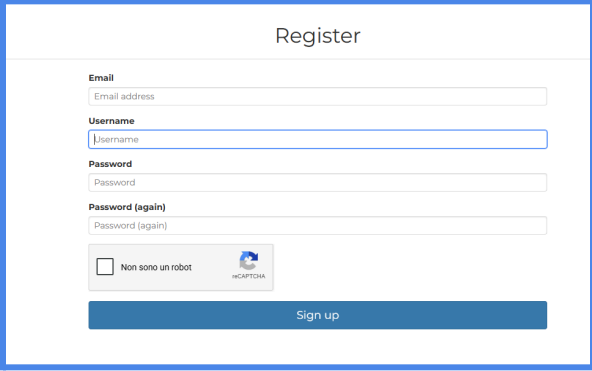
The image shows a web form titled "Register". It contains several input fields: "Email" with a sub-label "Email address", "Username", "Password", and "Password (again)" with a sub-label "Password (again)". Below these fields is a CAPTCHA section with a checkbox labeled "Non sono un robot" and a small image of a robot. At the bottom of the form is a blue button labeled "Sign up".

Figure. 2 Registration page

Therefore, registration requires an exchange of emails between the system administrator and the user. The system administrator's contact details are available on the "Data Policy" page, accessible via the link at the bottom of the homepage.

Once registered and approved, users can access the catalogue .using the “Sign in” link in the top of the home page

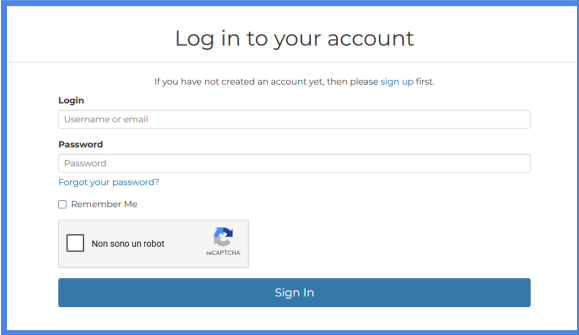
The image shows a web form titled "Log in to your account". It includes a link that says "If you have not created an account yet, then please sign up first." Below this is a "Login" section with input fields for "Username or email" and "Password". There is also a link for "Forgot your password?". A "Remember Me" checkbox is present. At the bottom is a CAPTCHA section with a checkbox labeled "Non sono un robot" and a small image of a robot. A blue button labeled "Sign In" is at the very bottom.

Figure 3 - Sign-in page

Users can access a resource only according to the permissions granted by the data managers, who are responsible for the data. Typically, the permissions granted to a generic user include the ability to view the data and its metadata.

The system's main page (Figure 1) displays the project logo in the upper-left corner, which redirects to the homepage when clicked.

In the upper-right corner there are: the language selection menu (English, Deutsch, Español, Français, Italiano); the user login icon provides access to the login page. Once the user is logged in, it displays a list of available actions, as shown in Figure 4:

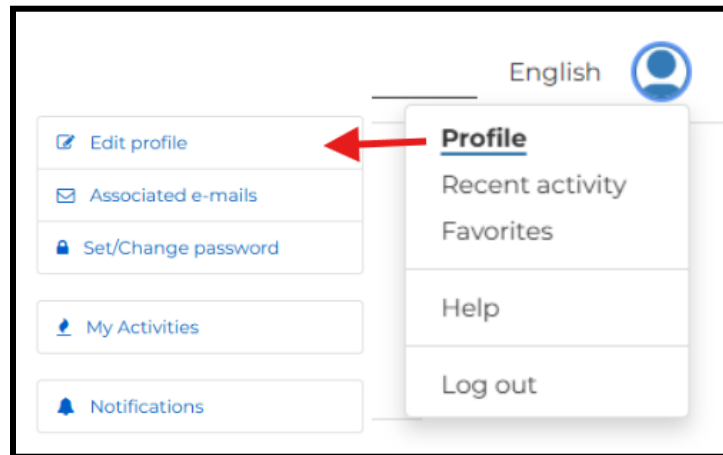


Figure 4 - User profile

- **Profile:** displays the user's profile
- **Recent activity:** displays the user's recent activities
- : displays the list of the favorites datasets
- : provides quick help on performing actions in the catalogue
- : logs the user out of the system.

In the top left, the menus provide direct access to the resources available to the user, organized by category (Datasets, Maps, Documents, Geostories, and Dashboards) or displayed together (All Resources). These resources can also be accessed by scrolling down the page.

In the center of the page header, there are quick links that provide direct access to the following resources, if made available by the data manager responsible for them:

- : Vector / Raster datasets with Pedological Classification
- : Point vector datasets with profile and samples soil data

- : Vector / Raster datasets with soil indicators describing chemical health
- : Vector / Raster datasets with soil indicators describing physical health
- : Vector / Raster datasets with soil indicators describing biological health



Figure 5 - List of resources and filters

The resource list can be filtered using the filter panel, accessible through the "Filter" button, as shown in Figure 5. When a filter is applied, a green dot is displayed on the "Filter" button.

From the list of available resources, clicking on the resource name opens the resource page, while clicking the three-dot menu on the right provides access to additional actions, including the preview. The preview provides an overview of the resource, including its geographic view, attributes, metadata, and other information made available for that resource (Figure 6).

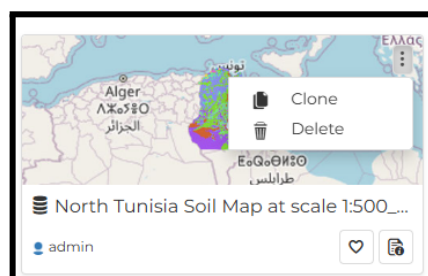


Figure 6 - the panel of a resource in the list

As mentioned above, the SIS resources can be distinguished in Datasets, Maps, Documents, Geostories, and Dashboards.

All or each resource can be filtered with one or more of the following filters:

- : resources owned by the user
- : resources explicitly marked as favorites by the user.
- : resources explicitly marked by data managers or administrators as highlighted or recommended
- that has been submitted and is awaiting approval by a data manager or administrator.
- : resources served by an external service, such as Web Map Service, ArcGIS REST service, or another GeoNode instance
- set of data
 -
 -
 -
 -
 -
- : all available documents
- : all available maps including those created using the Geonode Map Viewer, as well as maps created directly through MapStore-based viewer configuration files, through APIs or scripts, or imported from other Geonode instances
- : only maps created using the Geonode Map Viewer
- : all available Geostories
- : all available Dashboards
- : filters resources based on the value assigned to the Category metadata field. Typical categories, often based on INSPIRE / ISO topic categories used in geospatial metadata standards, include things like: Boundaries, Environment, Transportation, Economy, Health, Society, Utilities / Communication
- : filters resources based on the keywords assigned in the metadata, typically free text tags
- : filters resources based on metadata associated with the geographical region of the resource. This is distinct from the bounding box, which is automatically derived from the geographical extent of the data
- : filters resources based on the user who created or owns them
- : filters resources associated with a specific GeoNode group
- : filters resources based on their creation or upload date, showing only resources added on or after the specified date
 - : filters resources newer than the selected date
- : resources whose geographical area intersects with or is contained within the selected extent

Dataset are geospatial resources that can be stored, managed, visualized, and shared through the platform. A dataset typically represents a spatial layer.

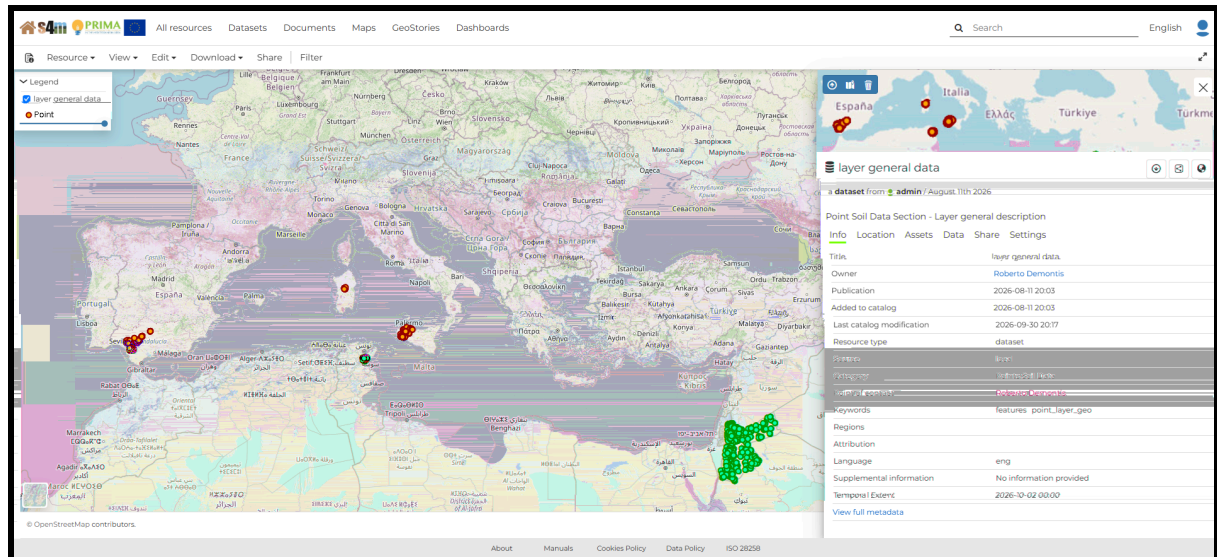


Figure 7: The dataset resource page

By clicking the (top left) the user can contemplate the resource's general information, geographical location, alphanumeric attributes, and the resources linked to the dataset, either as resources that use it or as resources used by it.

If the user has "manage" permission, the dataset can be saved or deleted through the

by clicking the , the user can access the resource's metadata and/or the alphanumeric data associated with the dataset. When clicking "view data", the advanced search icons in the upper left of the table provide the following options :

- Perform an advanced search using attribute filter, an area of interest, or a layer filter;
- filter the data displayed in the table according to the current view after zooming;
- zoom to the page extent;
- hide or show table columns;
- synchronize the map with the applied filters.

Examples of "advances search" functions are shown in Figure 8 and Figure 9

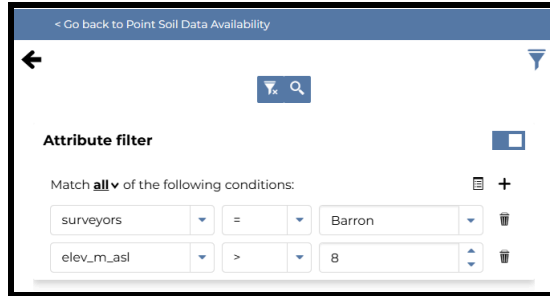


Figure 8

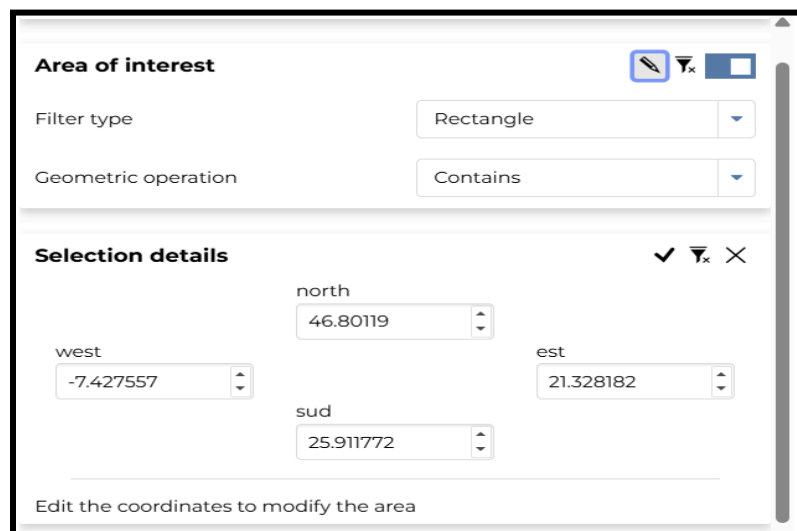


Figure 9

By clicking the , the user can:

- download the dataset as a compressed file containing the corresponding shapefile, which can be opened using GIS software;
- export the dataset in different formats including GeoJSON, GML-FeatureCollection, GML-GFS 1.1.0 FeatureCollection, Shapefile, and CSV;
- export only the portion of the dataset currently displayed in the map using Crop dataset to current viewport;
- Export only the subset defined by the applied filters using Download filtered dataset
- download the ISO metadata in .xml format ().

The “viewport” filter types the search results to resources that fall within the current map extent i.e., the area current visible on the screen.

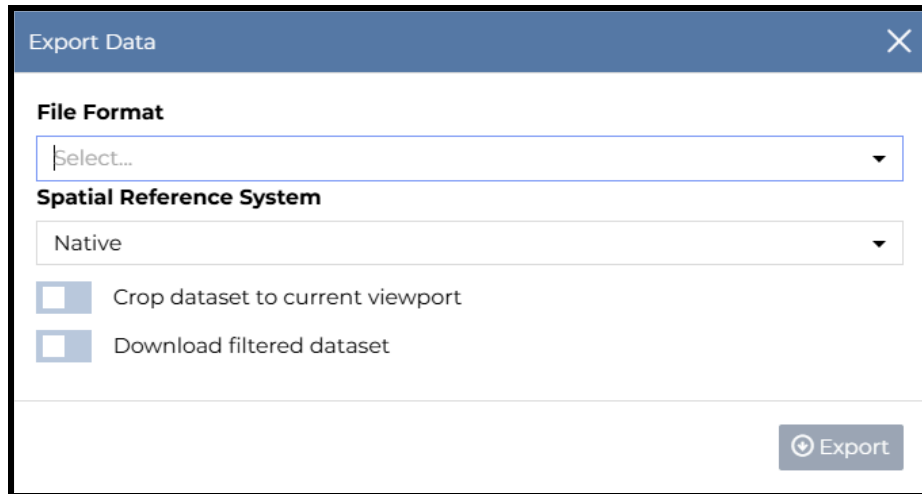


Figure 10

By clicking the [icon] the user can perform the same filtering operations available through the icons in the [icon]. The user can perform advanced searches including attribute-based filtering, and area of interest selection. For attribute filtering, multiple conditions can be applied by clicking the “+” button to add additional conditions. (Figure 11)

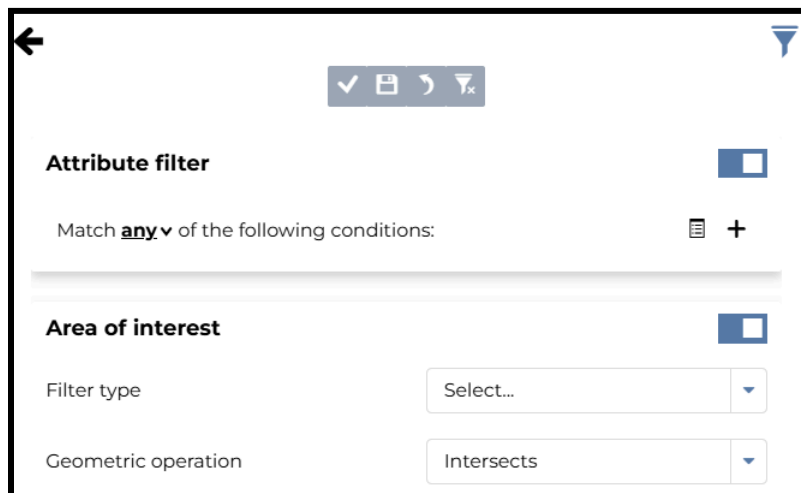


Figure 11

Maps are compositions or visualizations built from one or more datasets (layers).
The Figure 11 shows the map resource panel

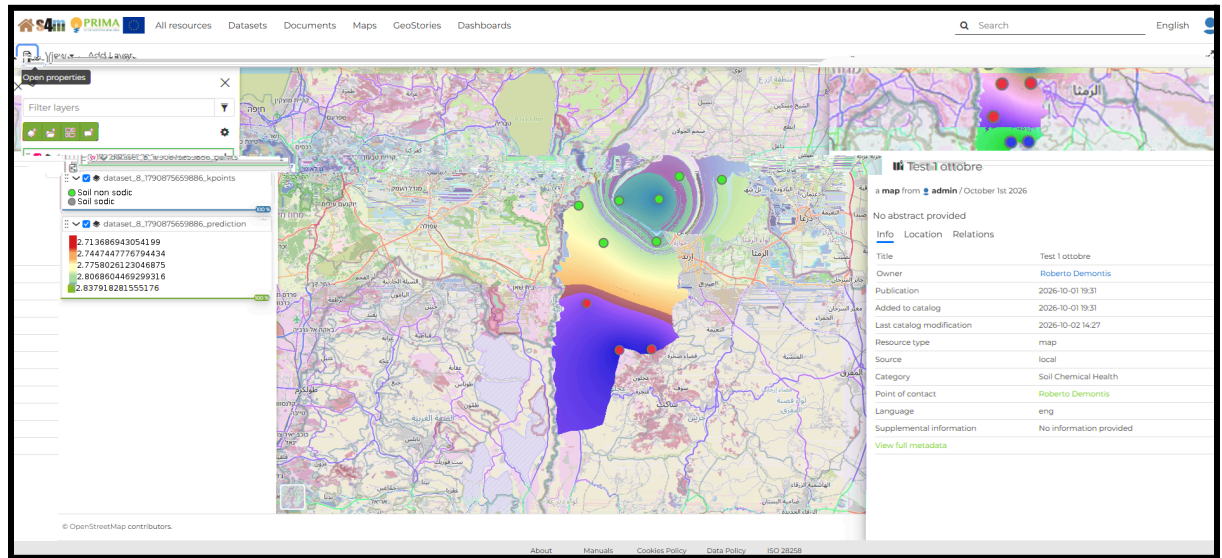
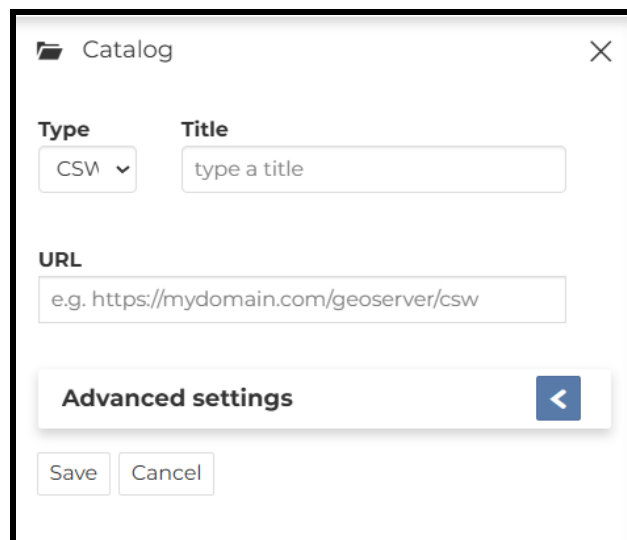
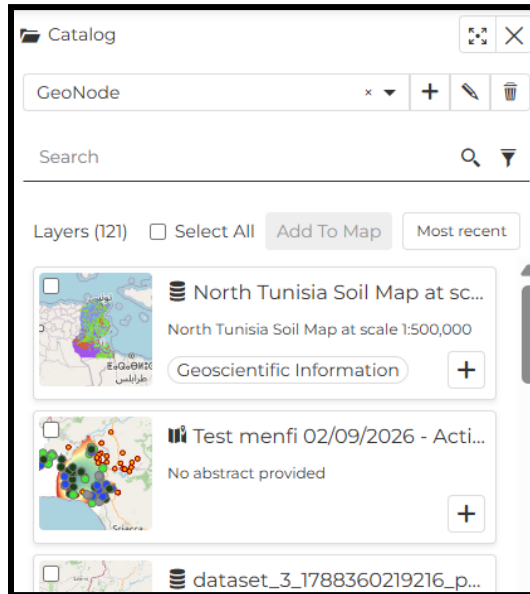
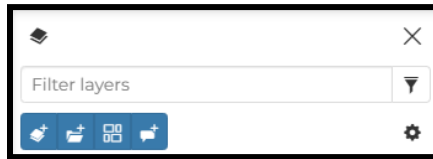


Figure 12: the map resource page

From the map resource panel users can:

- Access and view the map through the web interface
- Interact with the map by zooming, panning, toggling layers on or off
- Select a base map which is set to OpenStreetMap by default
- View the associated metadata (title, description, etc.) by clicking the
- Explore the layers included in the map by turning them on/off and viewing basic layer information. In some cases, when the layers included in the map are downloadable, user can also open the attribute table, query features, and apply filters
- Open the map directly through its URL
- Share the map with others by sharing its URL. Since the map is public, the link can be accessed by others. The sharing button is located on the right-hand side of the map information panel
- Add layers by clicking the first of the four buttons on the left (Figure 12, 13 and 14). User can add a layer of the catalogue or an external layer by selecting one of the following services or formats: WMS (Web Map Service), WMTS (Web Map Tile Service), CSW (Catalog Service for the Web), TMS (Tile Map Service), XYZ Tiles, WFS

(Web Feature Service), IFC Model, and ArcGIS REST.



The Advanced settings allow users to configure how a remote layer is:

- : service search: automatically retrieves available layers from the selected remote service, server type: defines the type of remote server (e.g., GeoServer, ArcGIS) to ensure the appropriate requests are sent
- : preview: displays a preview of the layer before it is adding it), format: sets the image format (e.g., PNG, JPEG) which affects quality and transparency, single tile vs tiled: determines whether the map is requested as a single large image or as a multiple tiles. Single tile mode may provide better image quality, but can result in slower performance, tile size: sets the size of the tiles (e.g., 256×256 pixels), affecting performance and the number of requests
- : visibility limits: determines the zoom levels at which the layer is visible, tile grids uses the tiling scheme defined by the remote service to ensure better alignment, domain aliases: specifies alternative domains for the same service which can be useful for load balancing or managing browser connection limits
- : interactive legend: allows users to interact with the legend, for example by toggling styles or classes, information sheet: defines the format of the feature information displayed when user clicks on the map, (e.g., HTML, JSON).

By clicking the second of the four buttons in Figure 12, users can add a new group of features to the current map.

By clicking the third and the fourth button shown in Figure 12, users can add widgets and annotations to the current map. If the user has edit permissions, annotations become part of the map. For a generic user without edit permissions, annotations are usually temporary and are lost when the page is refreshed.

Documents are used to upload, store and share non-spatial files through the platform, such as PDFs (reports, studies), Word documents, Excel spreadsheets, Images, Presentations.

Users may be able to open, download, edit, or manage documents. For example, a PDF can be previewed and downloaded (depending on permissions). As with other SIS resources, documents may have different access and management options depending on the user's permissions. Generic users can usually:

- the document and its metadata
- the document and its ISO metadata

The SIS includes documents related to Soil Point Data, Soil Maps, Soil Indicators and the User Manuals.

GeoStories are a special content type used to create interactive narratives that combine maps and multimedia content. They are essentially web-based storytelling pages that can integrate: maps already available in the SIS, text such as descriptions, and explanations, images and other media, embedded content.

If the GeoStory is public, a registered, non-privileged user can:

- the story and its metadata while following the narrative flow,
- with embedded maps to explore geographic content
- the GeoStory via its URL using the sharing button located on the right hand side

Dashboards are interactive pages used to visualize and analyze geospatial data through charts, widgets, and maps. They are data visualization tools that combine different elements, including: Maps, Charts (bar, pie, line, etc.), Indicators /counters, Filters and widgets. These elements allow users to summarize and explore data in a dynamic and visual way.

If the user has the necessary permissions to access the resources used in the dashboard, they can:

- the dashboard and its metadata including charts, maps, and widgets
- with the visualizations, for example by exploring chart bars, maps
- such as time-based or attribute-based filters, and see the visualizations update dynamically.

To access the soil data, users must be registered in the system and submit a request to the data managers—specifically, the SIS staff—using the contact details provided on the "Data Policy" page shown in Figure 15.

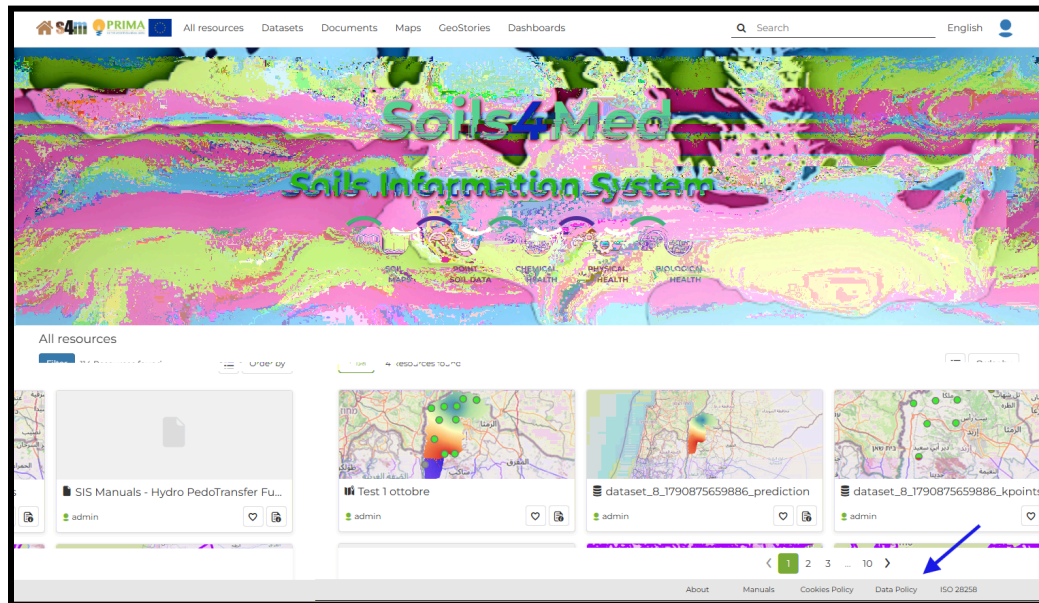


Figure 16

The request must include

- the `soil` parameter, which can also be specified using the name of an administrative division
 - A Soil indicator dataset resource name (see the document “MED-SIS-Manual-Soil_Indicators.pdf” in the Help page) or a set of map names;
 - A Point Soil data section dataset resource name (see the document “MED-SIS-Manual-Points_Soil_Data.pdf” in the Help page) or a set of names of map names
 - A set of Soil indicators and Point Soil data section datasets to be used to create a map
- The `soil` parameter, which can also be specified using the name of an administrative division

The request may also include the following parameters:

- start and end dates or all available dates
- upper and lower depth limits or all available depths
- : Auger hole, Minipit or Profile or all types
- field survey : the method used to carry out the field survey or all methods:

Identifier	Code	Description
SURVEY_METHODS:CP	CP	method: CPCS;
SURVEY_METHODS:FA	FA	method: FAO
SURVEY_METHODS:FA06	FA06	method: FAO; year: 2006
SURVEY_METHODS:FA52	FA52	method: FAO; year: 1952
SURVEY_METHODS:FA77	FA77	method: FAO; year: 1977
SURVEY_METHODS:FA90	FA90	method: FAO; year: 1990
SURVEY_METHODS:US	US	method: USDA soil survey manual
SURVEY_METHODS:US18	US18	method: USDA soil survey manual ; year: 2018
SURVEY_METHODS:US52	US52	method: USDA soil survey manual ; year: 1952
SURVEY_METHODS:US62	US62	method: USDA soil survey manual ; year: 1962
SURVEY_METHODS:US75	US75	method: USDA soil survey manual ; year: 1975
SURVEY_METHODS:US93	US93	method: USDA soil survey manual ; year: 1993
SURVEY_METHODS:WR	WR	method: WRB;
SURVEY_METHODS:WR22	WR22	method: WRB; year: 2022

- the Identifier or name of the project, or survey activity/campaign, within which the point soil data were generated or all projects
- laboratory the method used for the laboratory measurements, no specific method or all available methods, depending on the measurements involved.

If the requested dataset concerns a soil indicator, the user can also request an interpolation of the measurements over the selected area of interest. The interpolation will be performed using ordinary kriging.

After the requested datasets have been created and ,
the datasets:

- filtered point data,
- Interpolated data
- aggregated point data used for the interpolation

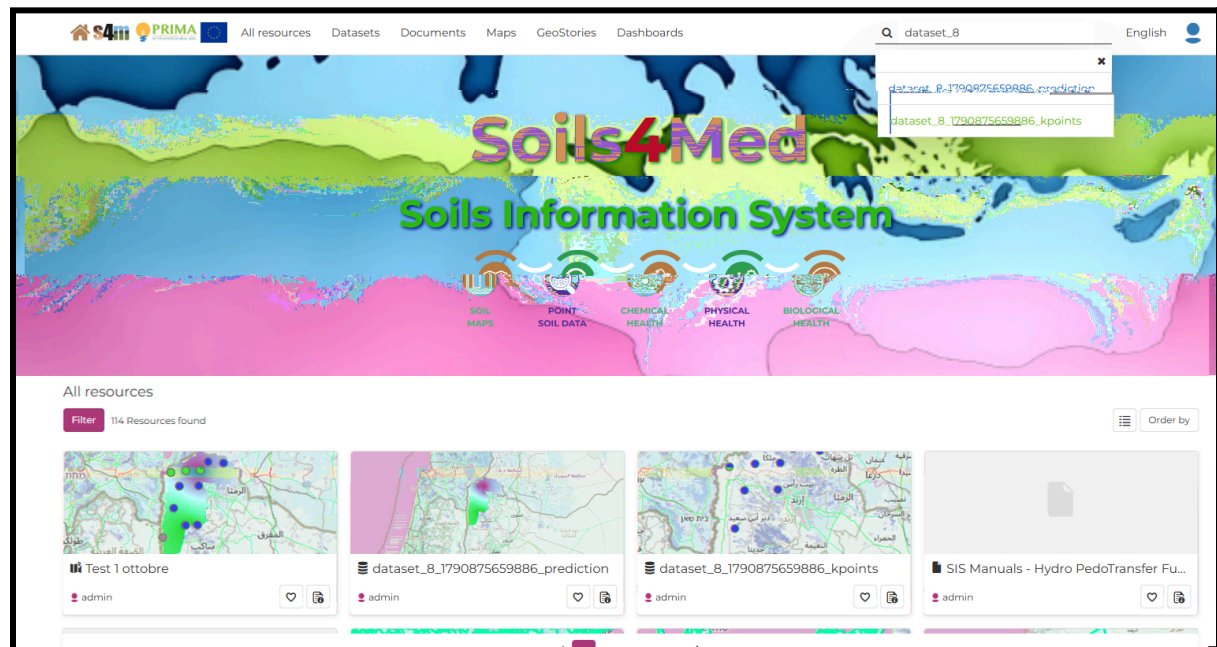


Figure 17